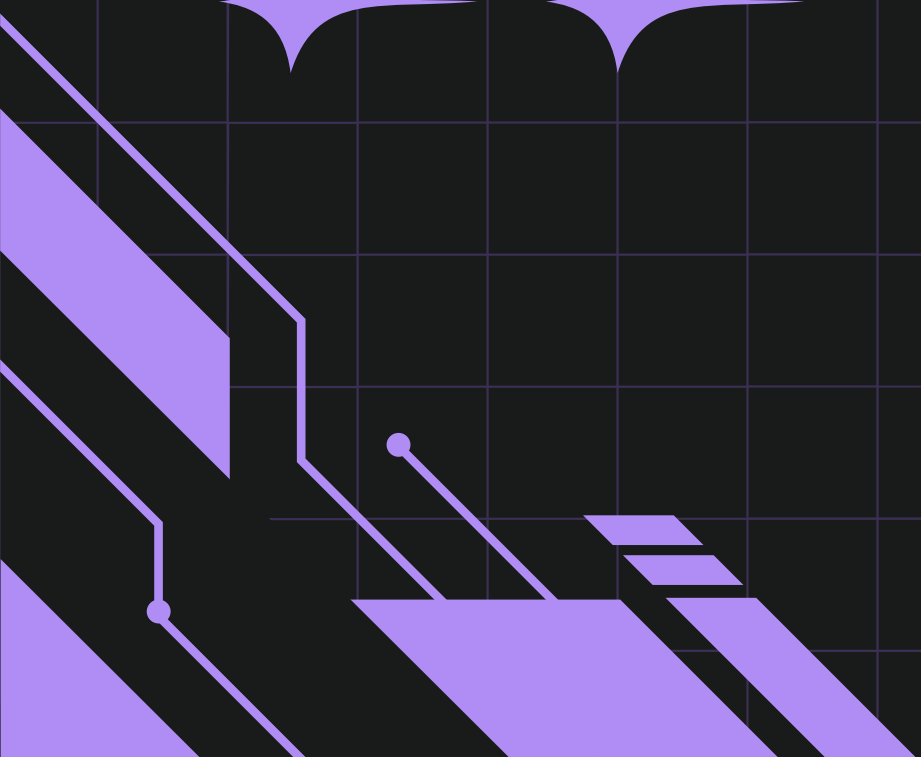
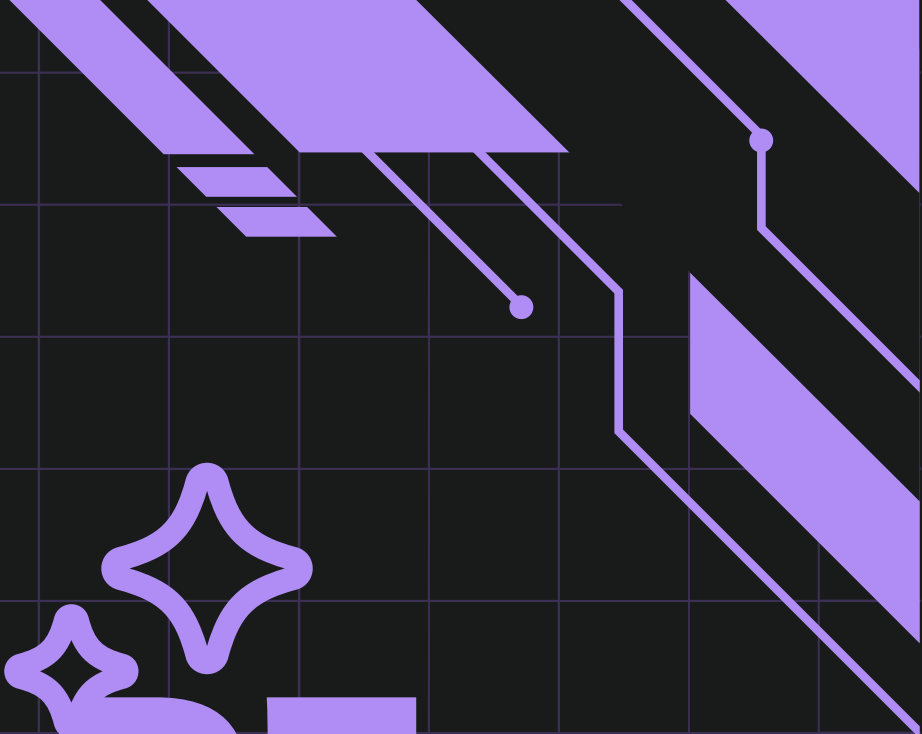
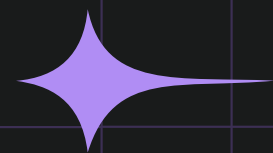




PARTE 1: SISTEMA DE ARQUIVOS BTRFS

SISTEMAS OPERACIONAIS

João Victor, Leonardo, Ludmila e
Maria Lissa



INTRODUÇÃO

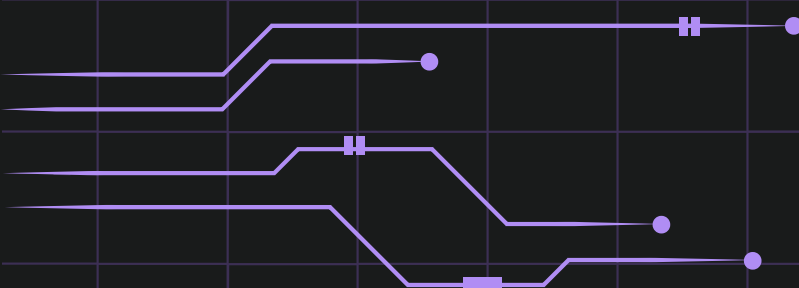
B-tree File System

estrutura de dados: árvores B

sistema de arquivos e
gerenciador de volumes

foca em tolerância a falhas,
recuperação e facilidade de
administração

btrfs

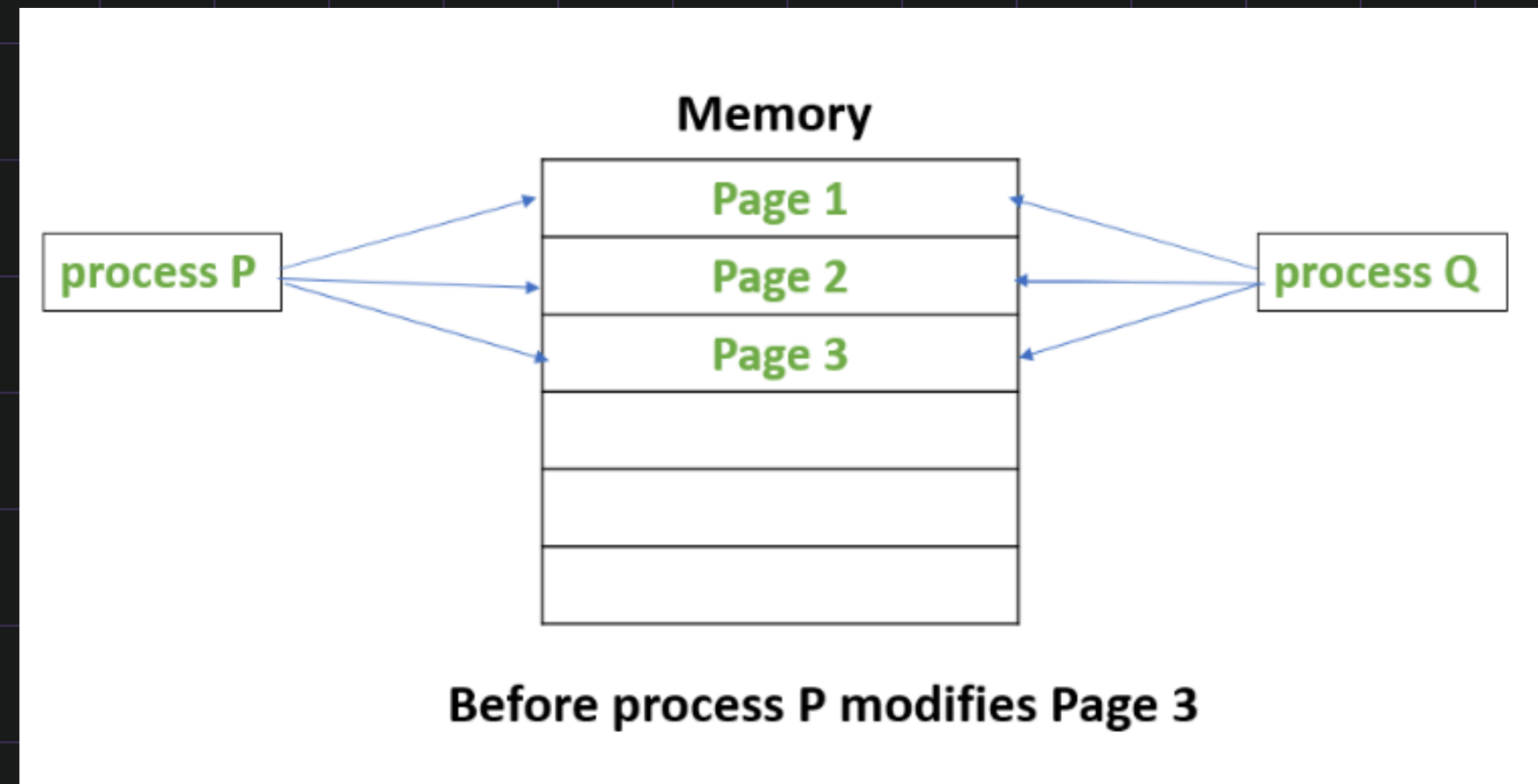


COPY ON WRITE

adia a cópia de um bloco de dados até que seja realmente necessário modificá-lo

cópia é somente feita na hora da escrita

evita operações de escrita desnecessárias



SUBVOLUMES E SNAPSHOTS

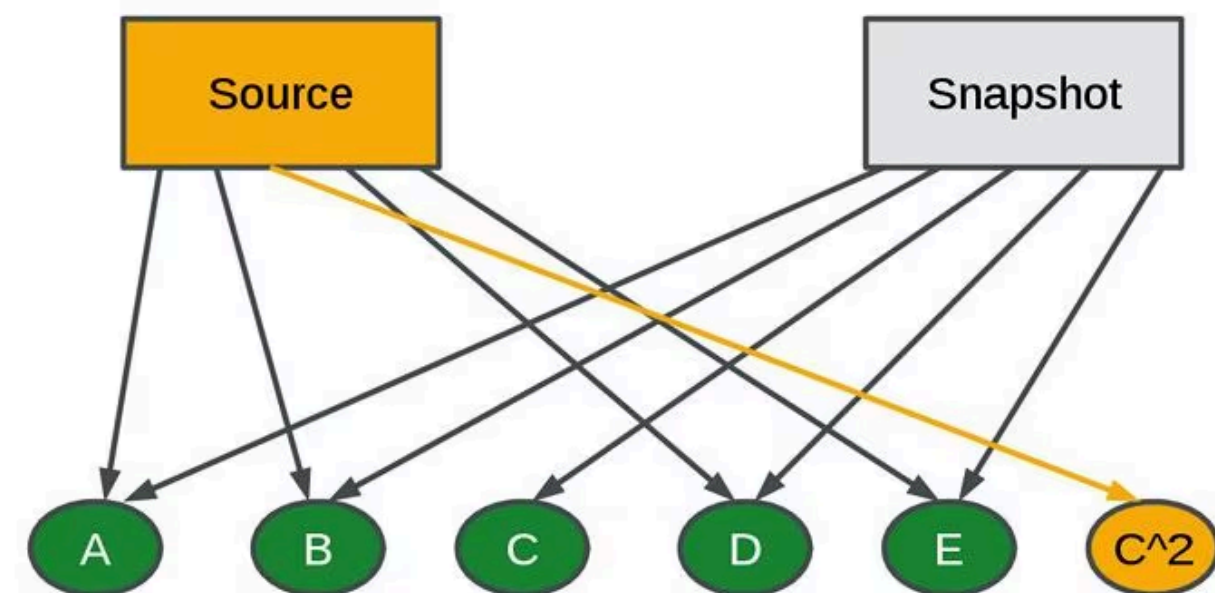
subvolumes são divisões lógicas dentro do sistema de arquivos

snapshot é um subvolume com conteúdo inicial

pontos de restauração

snapshots não consomem muito espaço

Btrfs / Copy-on-Write (4/4)



técnica utilizada para verificar integridade de dados.

algoritmo CRC

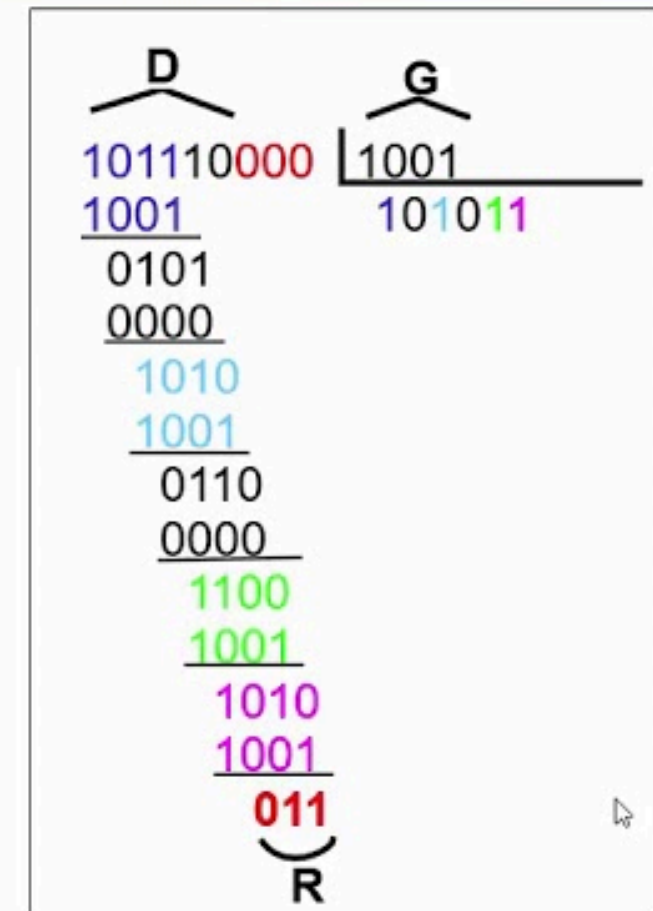
Checksum Tree

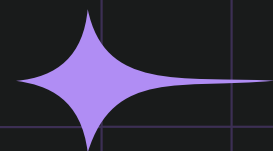
- D = 101110 (datos)
- G = 1001
 - r (el CRC) tendrá 3 bits

- Añade r ceros al final de D
- Realiza la división
- Envía D+R:

- Divide los datos recibidos (D+R) por el G acordado

- Si el resto = 0 $\rightarrow$ ¡Datos correctos!





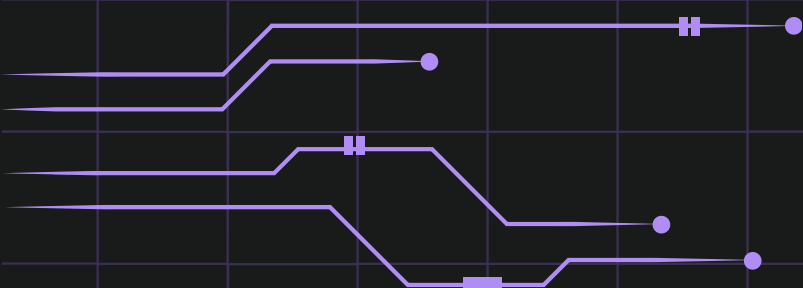
OUTRAS FUNCIONALIDADES

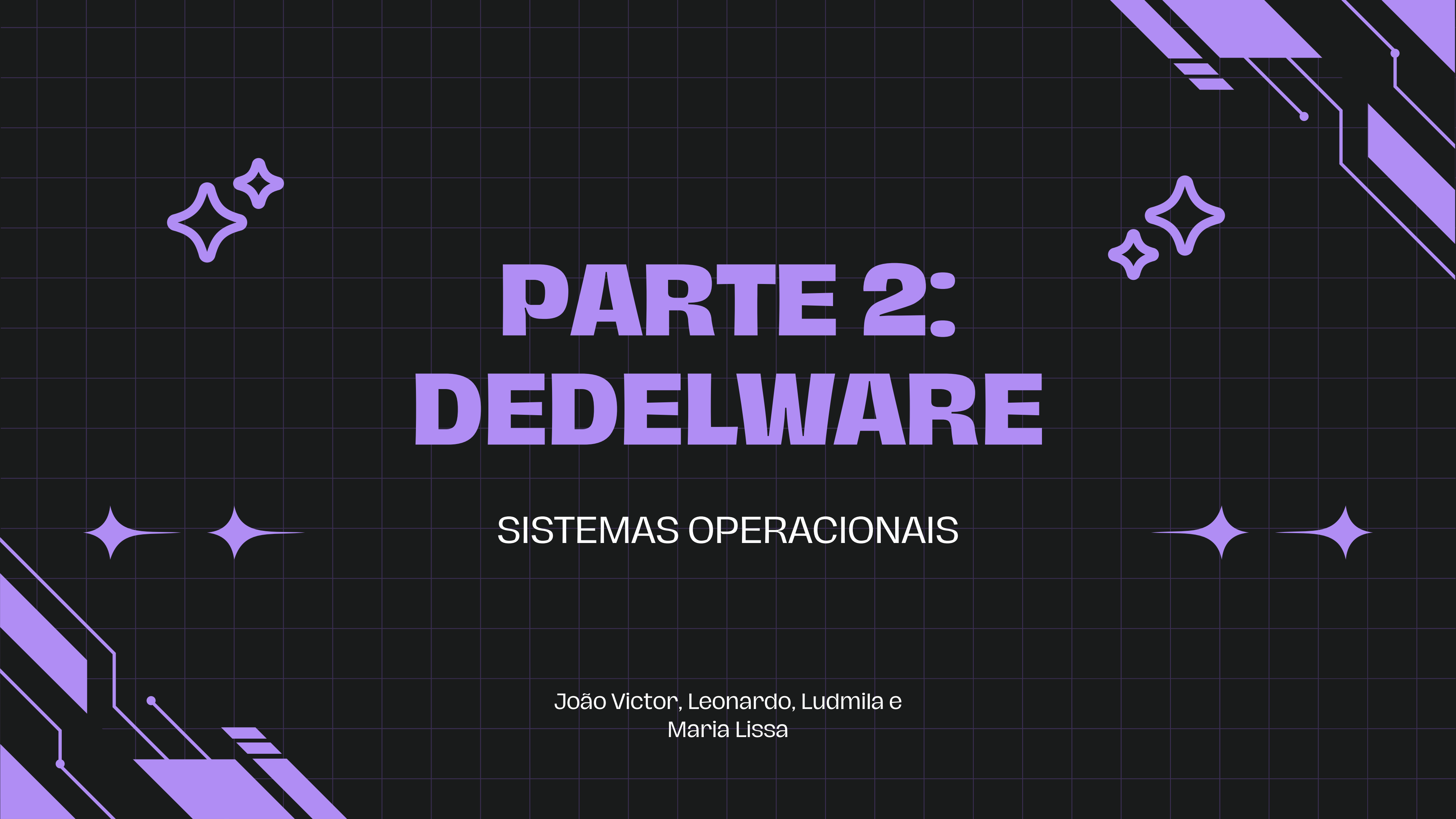
Compressão

Gerenciamento de discos e
RAID

Send e Receive

Três cópias de Superblock
(64Kib, 256MiB e 256GiB)

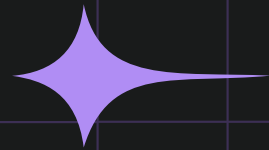




PARTE 2: **DEDELWARE**

SISTEMAS OPERACIONAIS

João Victor, Leonardo, Ludmila e
Maria Lissa



DEDEL
WARE

Anatomia de um

SHELL REVERSO

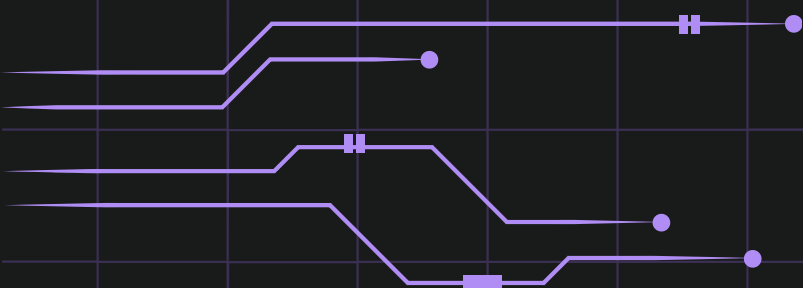
Malware por Engenharia Social

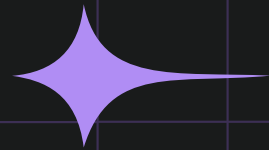
Geração de Processos

Redirecionamento de Fluxos

Chamadas de Sistema

Linguagem C





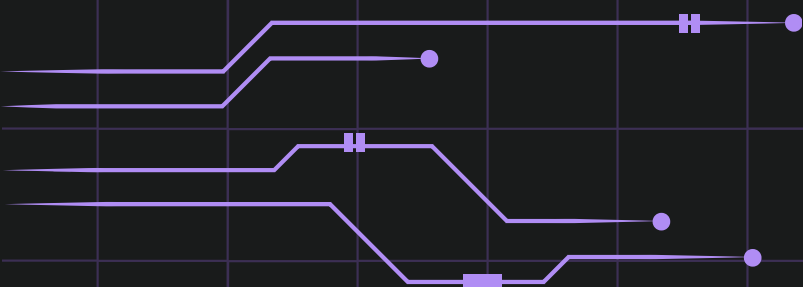
DEDEL
WARE

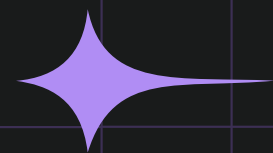
MALWARE ENG. SOCIAL

Servidor em Escuta Ativa

Vítima Voluntária

Contorna Firewall





GERAÇÃO DE PROCESSOS

Processo Pai: id de processo A

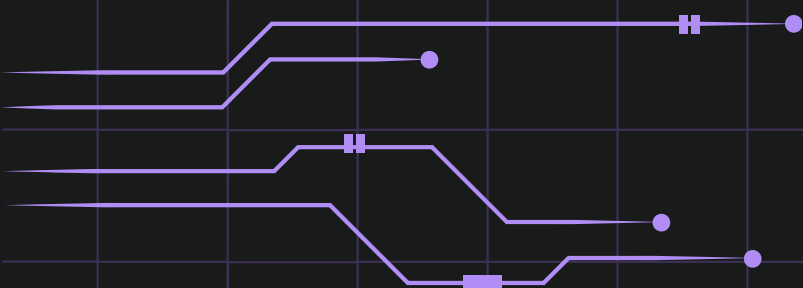
pid $\leftarrow$ 0

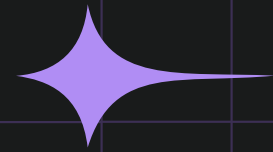
fork() $\rightarrow$ **Processo pai:** id de processo B

pid $\leftarrow$ id de processo B

pid $\leftarrow$ 0

Estabelece conexão ...





REDIRE CIONA MENTO

Processo Pai: id de processo A

pid $\leftarrow$ 0

fork() $\rightarrow$ **Processo filho:** id de processo B

pid $\leftarrow$ id de processo B

pid $\leftarrow$ 0

Estabelece conexão ...

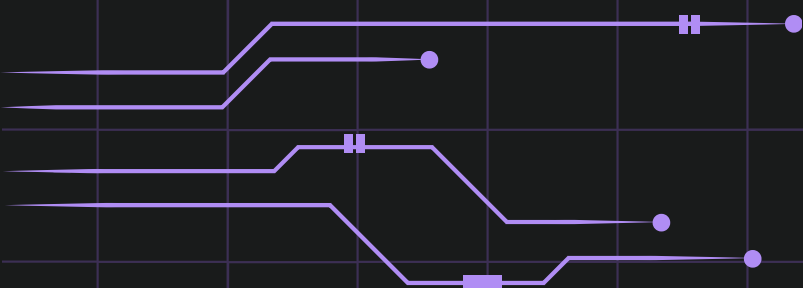
dup2

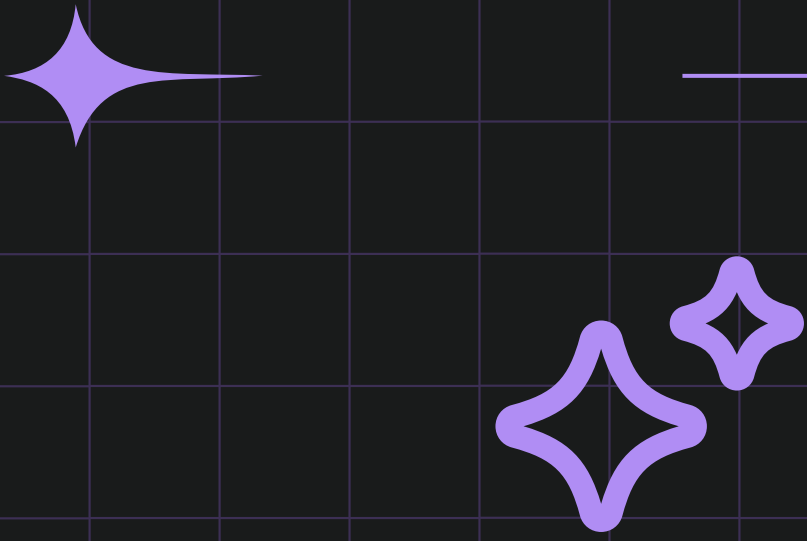
socket $\leftarrow$ 0 “e”

socket $\leftarrow$ 1 “s”

socket $\leftarrow$ 2 “derr”

execve() $\rightarrow$ Abre o Shell





CÓDIGO VÍTIMA

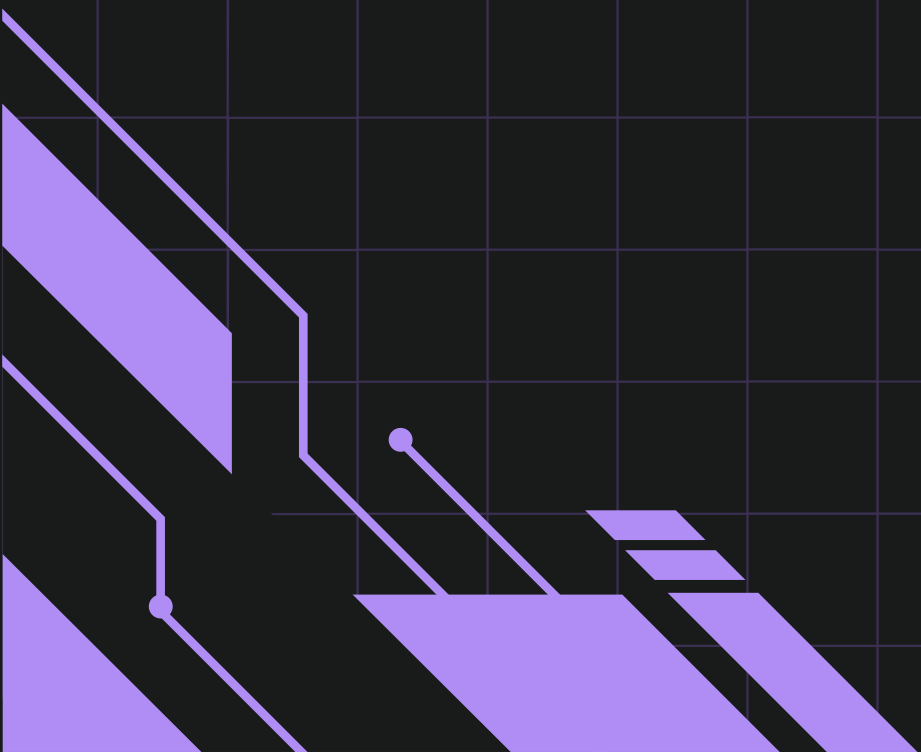
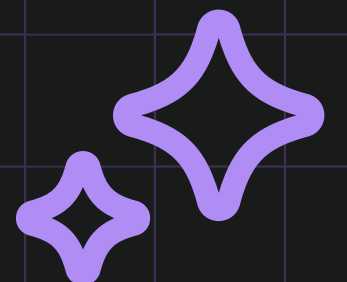
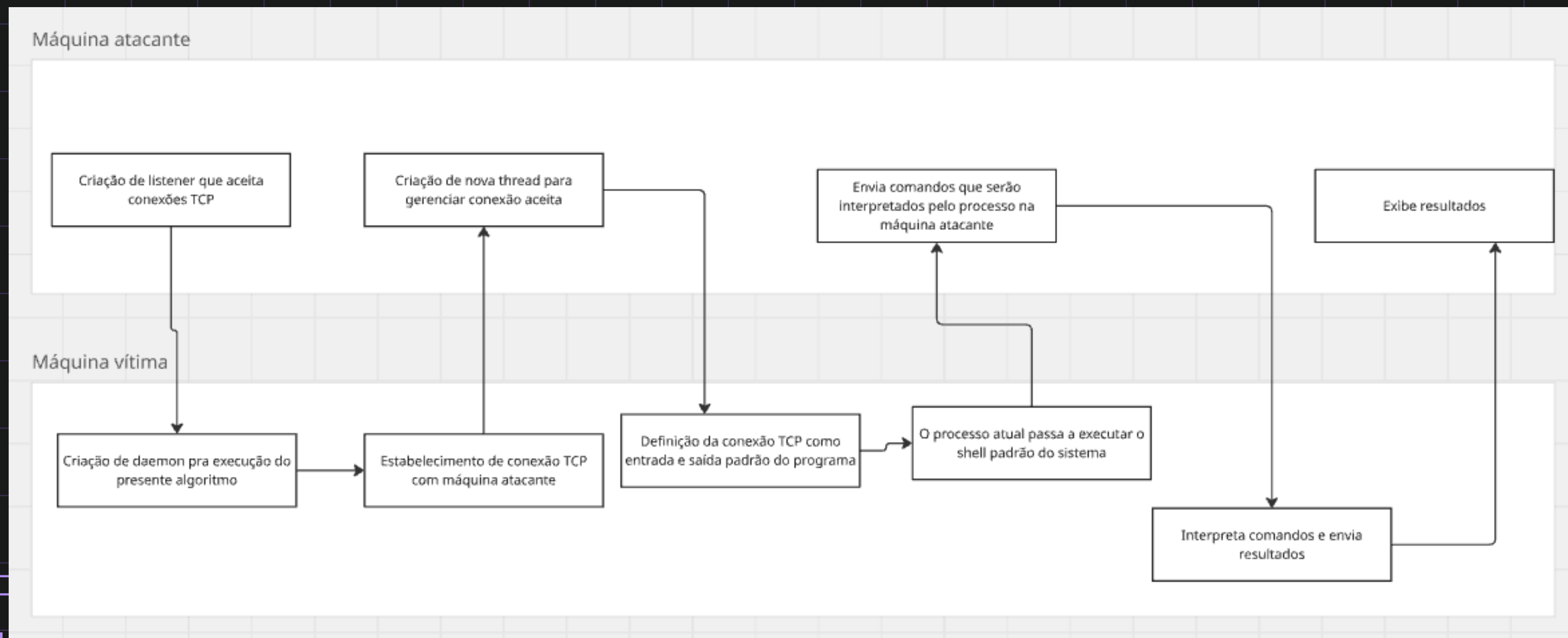
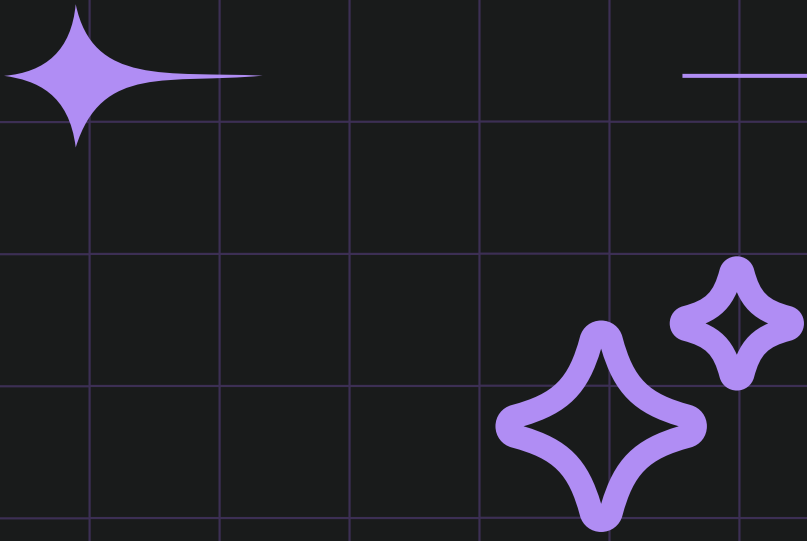
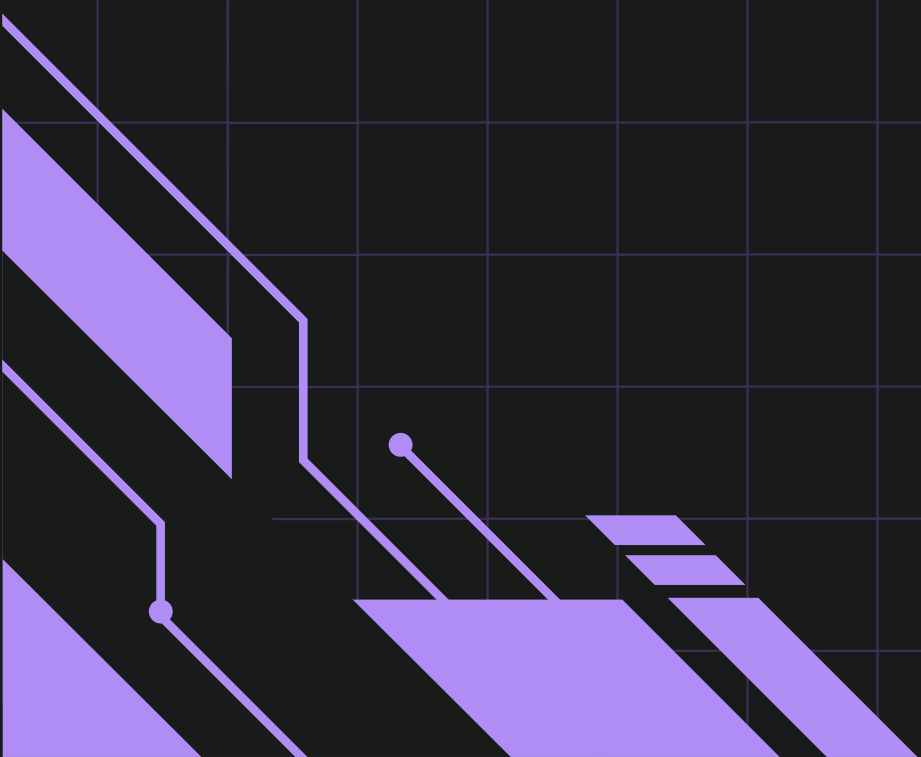
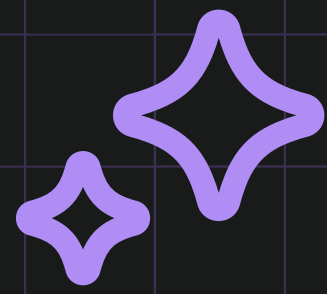


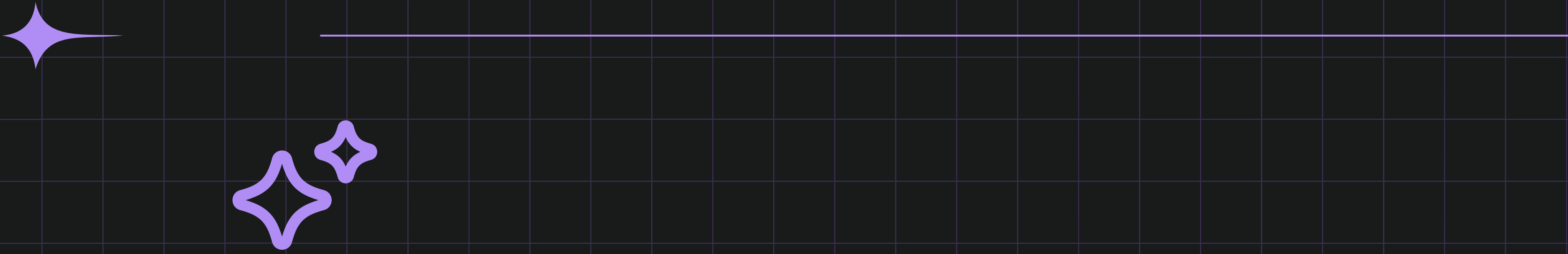
DIAGRAMA DE BLOCOS





CÓDIGO ATAQUE





OBRIGADO!

